

VRDots — Data Files Guide

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Overview

Every experimental session produces **three files**, all named with the session timestamp:

```
vr_dots_session_YYMMDD_HHMM.tsv
vr_dots_session_YYMMDD_HHMM.tsv.meta.json
vr_dots_session_YYMMDD_HHMM.tsv.sidecar.json
```

Send all three files to Gene after every session. The `.tsv` contains the trial-by-trial results. The `.sidecar.json` contains the complete stimulus parameters. The `.meta.json` is a brief session summary.

The trial data file (`.tsv`)

A tab-separated table. One row per completed trial, one column per variable. Open in Excel or any spreadsheet app.

Key columns

Column	What it means
Trial	Trial number within the session (starts at 0)
Experiment	Internal experiment identifier (e.g. DensityCompare_HighDens_v2)
Cond	Whether the translating field appeared later (CUED) or earlier (UNCUED)
TransDeg	True translation direction in degrees (0°=right, 90°=up, 180°=left, 270°=down, and four diagonals)
RespDeg	Observer's reported direction in degrees
SwapType	What swapped at translation onset: N=nothing, CM=colors+motion directions both swap
RTf	Response time in milliseconds from translation onset
DelayedFieldColor	Which color field appeared later: R=red, G=green
SeedA0–SeedB3	Random seeds used to generate this trial's dot positions (enables exact replay)
MkHash32	Fingerprint of the motion trajectory — use to verify trial identity across sessions

Computing accuracy

A trial is **correct** when `TransDeg == RespDeg` .

```
import csv
```

```

with open('vr_dots_session_YYMMDD_HHMM.tsv') as f:
    trials = list(csv.DictReader(f, delimiter='\t'))

cued = [t for t in trials if t['Cond'] == 'CUED']
uncued = [t for t in trials if t['Cond'] == 'UNCUED']

def pct(rows):
    correct = sum(1 for t in rows if float(t['TransDeg']) == float(t['RespDeg']))
    return 100 * correct / len(rows) if rows else 0

print(f"CUED: {pct(cued):.1f}% (n={len(cued)})")
print(f"UNCUED: {pct(uncued):.1f}% (n={len(uncued)})")
print(f"Cueing effect: {pct(cued) - pct(uncued):+.1f} pp")

```

Chance performance is **12.5%** (1 of 8 equally-spaced directions). A healthy session typically shows CUED ~60–70% and UNCUEd ~25–35%, giving a cueing effect of +25–35 percentage points.

The sidecar file (`.tsv.sidecar.json`)

The authoritative record of **exactly what stimulus was shown**. Always keep this alongside the `.tsv`. The most important section is `experiment_spec`.

`experiment_spec` — the stimulus parameters

Every parameter that defines the stimulus is recorded here. Key fields:

Field	Typical value	What it means
<code>spec_name</code>	<code>Exp_DensityCompare_HighDens</code>	Name of the experiment asset loaded
<code>dot_size_deg</code>	<code>0.12</code>	Dot diameter in degrees of visual angle
<code>dots_per_field</code>	<code>173</code>	Dots per perceptual field (red or green)
<code>aperture_radius_deg</code>	<code>3.5</code>	Radius of the circular dot aperture in degrees
<code>view_distance_m</code>	<code>2.0</code>	Virtual viewing distance in metres
<code>sim_hz</code>	<code>90</code>	Simulation frame rate
<code>rotation_speed_deg_per_sec</code>	<code>81</code>	Background rotation speed
<code>translation_speed_deg_per_sec</code>	<code>2.26</code>	Translation speed
<code>translation_duration_ms</code>	<code>80</code>	How long translation lasts
<code>delayed_onset_ms</code>	<code>750</code>	Delay between field onsets (the cue)
<code>repeats_per_stimulus</code>	<code>8</code>	Trials per unique stimulus condition

include_no_swap_baseline	true	Whether N (no-swap) trials are included
include_cm_swaps	true/false	Whether CM (color+motion swap) trials are included
fixation_exclusion_radius_deg	1.1	Radius of the dot-free zone around fixation

calibration_colors — dot colors

```
"calibration_colors": {
  "rgba_red": [0.8, 0.2, 0.2, 1],
  "rgba_green": [0.133, 0.545, 0.133, 1]
}
```

These are the RGBA values used for red and green dots. They are set to approximate equal luminance (verified by flicker photometry).

stimulus_builder — rendering method

Records how dots were rendered on screen:

Field	Value	Meaning
use_screen_space_shader	true	Dots drawn as flat quads (not 3-D spheres)
use_fixed_aa_shader	true	Fixed anti-aliasing — stable dot brightness
dot_blend_mode	additive	Color overlap adds; no occlusion between red/green
respawn_when_out_of_bounds	true	Dots that leave the aperture reappear inside it

Important: if `use_fixed_aa_shader` is `false`, dot brightness may vary slightly with sub-pixel position. All sessions from 2026-09-16 onward use the fixed shader.

build_date

The date and time the app was built (e.g. `2026-09-16 12:37`). Use this to identify which version of the code was running if there is ever a question about stimulus correctness.

The meta file (`.tsv.meta.json`)

A brief session summary. Useful for a quick sanity check:

```
{
  "experiment_name": "DensityCompare_HighDens_v2",
  "stats": {
    "target_number_trials": 512,
    "completed_trials": 513
  }
}
```

```
}  
}
```

`completed_trials` may exceed `target_number_trials` by 1 because a trial that is re-queued (observer re-donned headset mid-session) adds one extra completion.

Pulling data from the headset

With the Quest 3 connected via USB and USB debugging enabled:

```
adb pull /sdcard/Android/data/com.genestoner.vrdptsrebuildX.test/files/  
~/VRDotsData/
```

This copies all session files to a folder called `VRDotsData` in your home directory. Send the three files for each session to Gene at `generstoner@gmail.com`.

File naming

`vr_dots_session_260916_1240.tsv` = session recorded on **2026-09-16 at 12:40**.

The timestamp is set by the Quest's internal clock when the session begins. If the Quest clock is wrong, timestamps will be wrong — check the clock in Quest Settings if dates look implausible.

Contact

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